Introduction to Logic: Lecture 3

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What this course is about (**reminder**)...

Logic: the study of reasoning, inference, and how one constructs correct arguments.

An argument in English

"You can access the Internet from campus only if **you are a linguistics major** or you are **not a freshman**. Mary **can access the Internet**. So, Mary is **a linguistics major** or **not a freshman**."

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Today we will introduce classical (propositional) logic

$$P_a \rightarrow (P_l \vee \neg P_f), \quad P_a \quad \therefore \quad P_l \vee \neg P_f$$

Tools we will learn along the way and **where we are**

Understanding and building logics requires a few mathematical tools:

$\{a, b, c\}$

Sets

linguistics majors

$R(a, b)$

Relations

loves, older than

$f : A \to B$

Functions

capital of

$\vdash$

Proofs

informal & formal

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linguistics majors	*loves, older than*	*capital of*	informal & formal
first lecture	last lecture	last lecture	

Last lecture: relations and functions

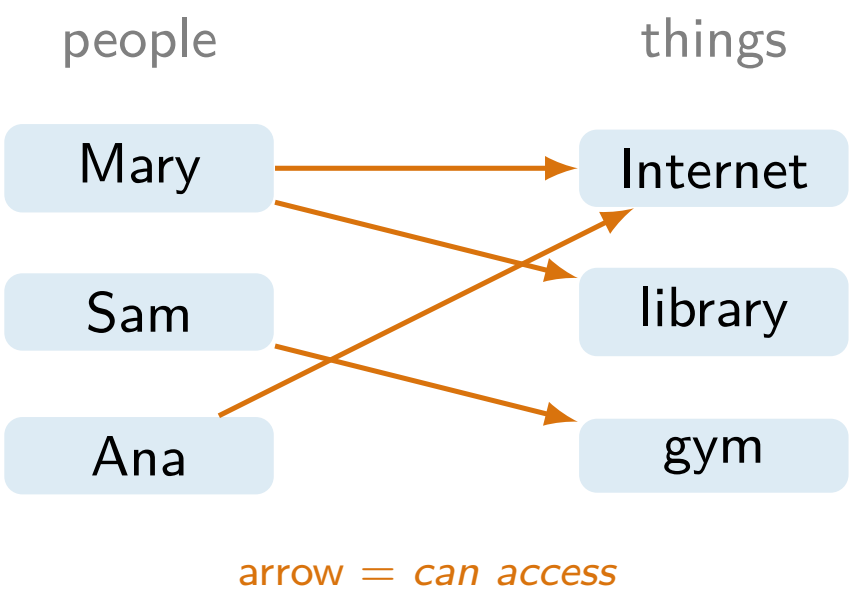
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Who can access what?

$\textsc{Access}(\text{Mary}, \text{Internet})$ ✓

$\textsc{Access}(\text{Sam}, \text{Internet})$ ×

$\textsc{Access}(\text{Ana}, \text{Internet})$ ✓

Relations describe how things are *connected*, we showed how they can be formalized as types of sets; functions are particular types of relations.

Today: Proofs and Propositional Logic

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- Start turning what we learned into formal statements and proofs; introduce classical propositional logic.

Reminders from the last lecture

Relations and functions: what you should know

Tuples and Cartesian products

- ✓ $\langle a, b\rangle \neq \langle b, a\rangle$: order matters
- ✓ $X \times Y$ and $|X \times Y| = |X| \cdot |Y|$

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Relations

- ✓ $R \subseteq X \times Y$, written Rxy
- ✓ read and draw arrow diagrams
- ✓ relations on one set; n-ary
- ✓ $\mathrm{dom}(R)$ and $\mathrm{range}(R)$

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- ✓ codomain Y vs. $\mathrm{range}(f)$

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Structural properties

- ✓ reflexive, symmetric, transitive, . . .
- ✓ injective, surjective, bijective
- ✓ refute with *one* counterexample

Quick check: tuples and relations

True or false?

1. $|\{a, b\} \times \emptyset| = 2$
2. $\langle a, \langle b, c\rangle\rangle = \langle\langle a, b\rangle, c\rangle$
3. there are 16 relations on $\{a, b\}$
4. every asymmetric relation is irreflexive
5. every symmetric and transitive relation is reflexive
6. $\text{dom}(R) = \text{range}(R)$ for every $R \subseteq X \times X$

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   first elements: $a \neq \langle a, b\rangle$

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These are again tricky!

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2. $\{\langle 1, a\rangle, \langle 1, b\rangle\}$ is a function $\{1\} \to \{a, b\}$
3. there are 9 functions $\{1, 2\} \to \{a, b, c\}$
4. some $f : \{1, 2\} \to \{a, b, c\}$ is surjective
5. $f(x) = 2x$ is a bijection $\mathbb{N} \to \mathbb{N}$
6. if X is finite, every injective $f : X \to X$ is surjective

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   true
   n inputs hit n distinct outputs

Questions?

Being more formal, proofs, infinity

We have casually stated many definitions and facts

Not everything we said was the same kind of statement.

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Definitions

what a word or symbol *means*: we choose them

- $X \subseteq Y$: each $x \in X$ is in Y
- $\emptyset$: the set with no elements
- R is reflexive: Rxx for all x

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Assumptions

taken for granted, *not* shown

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- $0 \in \mathbb{N}$
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- if $X \subseteq Y$ and $Y \subseteq X$, then $X = Y$
- $\mathbb{N}$ is infinite

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Definitions and assumptions we simply *accept*.
But why should we believe the **facts**?

Facts follow from definitions and assumptions

Things we said along the way

$\emptyset \subseteq X$ for every set X

there is only *one* empty set

if $X \subseteq Y$ and $Y \subseteq X$, then $X = Y$

$|\mathcal{P}(X)| = 2^{|X|}$ for every finite X

every asymmetric relation is irreflexive

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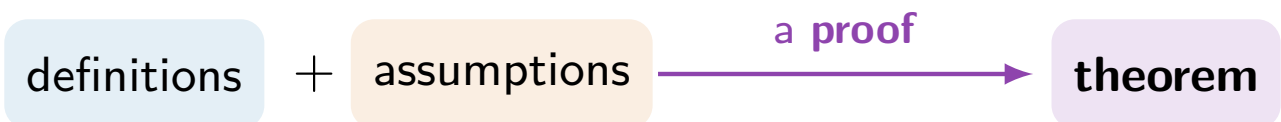
Things we said along the way

$\emptyset \subseteq X$ for every set X	$\leftarrow$ meaning of $\subseteq$ and $\emptyset$
there is only *one* empty set	$\leftarrow$ sets are fixed by their elements
if $X \subseteq Y$ and $Y \subseteq X$, then $X = Y$	$\leftarrow$ meaning of $\subseteq$ + same assumption
$\lvert\mathcal{P}(X)\rvert = 2^{\lvert X\rvert}$ for every finite X	$\leftarrow$ meaning of $\mathcal{P}$ + counting
every asymmetric relation is irreflexive	$\leftarrow$ meaning of both words

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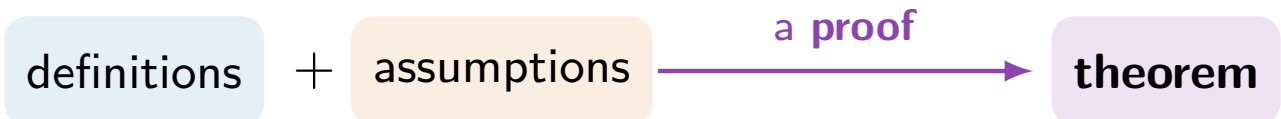
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every asymmetric relation is irreflexive	← meaning of both words



A statement shown to follow from definitions and assumptions is called a **theorem**; the argument that shows it is its **proof**.

Some important mathematical vocabulary

Words you will see

definition	what a word means
assumption	accepted without proof
theorem	a proven statement
proposition	a smaller theorem
lemma	a helper theorem
corollary	follows quickly
proof	the argument ■

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If $X \subseteq Y$ and $Y \subseteq X$, then $X = Y$.

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A proof is a chain of steps

each step follows from
the hypotheses, a definition,
or an earlier theorem

There are different styles of proof

Strategy 1: refutation by counterexample

Claim

For all a and all sets X, Y: if $X \in Y$ and $a \in X$, then $a \in Y$.

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Try an example

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$X \in Y$ ✓ $a \in X$ ✓ $a \in Y$ ✓

the claim holds in this case

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To refute a claim about *all* cases: *one* counterexample is enough.

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To prove it: examples are never enough, we need an argument that covers *every* case.

Strategy 2: proof by contradiction

The recipe (*reductio ad absurdum*)

1. **assume the opposite**: suppose a *counterexample* exists
2. reason until you reach a **contradiction**: something both true and false
3. conclude: *no* counterexample can exist, so the claim is **true**

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For every set X: $\emptyset \subseteq X$.

Proof

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By the definition of $\subseteq$, some element of $\emptyset$ is not in X.

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By the definition of $\subseteq$, some element of $\emptyset$ is not in X.
But $\emptyset$ has *no* elements. **Contradiction!** So $\emptyset \subseteq X$ for every X. ■

Two more proofs by contradiction

Proposition 2

There is only one empty set.

Proof

Assume the opposite: there are two different empty sets, $\emptyset_1 \neq \emptyset_2$.

Sets are determined by their elements, so some x is in one but not the other.

But then x is an element of an empty set, and neither has *any* elements.

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Contradiction! ■

Proposition 3

If $X \subseteq Y$ and $Y \subseteq X$, then $X = Y$.

Proof

Assume $X \subseteq Y$ and $Y \subseteq X$, and assume the opposite: $X \neq Y$.

Sets are determined by their elements, so some $x \in X$ is not in Y, *or* some $y \in Y$ is not in X.

The first breaks $X \subseteq Y$; the second breaks $Y \subseteq X$.

Contradiction! ■

Two more proofs by contradiction

Proposition 2

There is only one empty set.

Proof

Assume the opposite: there are two different empty sets, $\emptyset_1 \neq \emptyset_2$.

Sets are determined by their elements, so some x is in one but not the other.

But then x is an element of an empty set, and neither has *any* elements.

Contradiction! ■

Proposition 3

If $X \subseteq Y$ and $Y \subseteq X$, then $X = Y$.

Proof

Assume $X \subseteq Y$ and $Y \subseteq X$, and assume the opposite: $X \neq Y$.

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Proposition 3 is *how* we prove two sets are equal: show $\subseteq$ in both directions.

These are *informal* proofs

What the proof of Proposition 3 used

hypotheses $X \subseteq Y, \quad Y \subseteq X$

definition meaning of $\subseteq$

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- ▷ in ordinary language
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Informal proofs

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Formal proofs (later in the course) leave nothing out: every step is justified by an explicit rule, so even a computer can check it.

To get there, we first need a precise language for *not*, *and*, *or*, . . . : propositional logic.

A formal proof: Proposition 3 in the Lean proof assistant

```
-- Proposition 3: if X ⊆ Y and Y ⊆ X, then X = Y
theorem prop3 {α : Type} (X Y : Set α)
    (h₁ : X ⊆ Y) (h₂ : Y ⊆ X) : X = Y := by
  ext x              -- sets are fixed by their elements
  constructor        -- show both directions of ↔
  · intro hx         -- take any x ∈ X
    exact h₁ hx      -- X ⊆ Y gives x ∈ Y
  · intro hy         -- take any x ∈ Y
    exact h₂ hy      -- Y ⊆ X gives x ∈ X
```

`ext x` is the step our informal proof skipped: here it must be stated explicitly, and the computer *checks* every line.

Infinity

Finite and infinite sets

Definition

A set X is **finite** if and only if (iff) there is **no bijection** $f : X \to Y$ from X to a **proper subset** $Y \subset X$. Otherwise, X is **infinite**.

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Two words from before

bijection $f : X \to Y$ each x gets exactly one partner, none left over

proper subset $Y \subset X$ $Y \subseteq X$, but $Y \neq X$: something is left out

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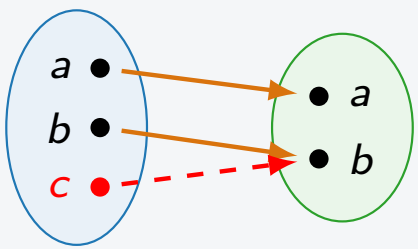
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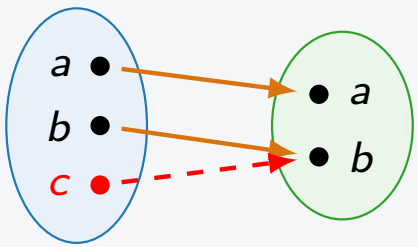
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Infinite: a part can be as big as the whole

?

sounds strange, but this is what happens with $\mathbb{N}$

A proof by contradiction: $\mathbb{N}$ is infinite

Proposition 4

The set of natural numbers $\mathbb{N} = \{0, 1, 2, \ldots\}$ is infinite.

Proof

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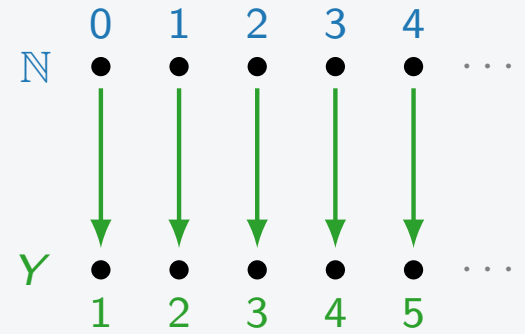
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Take $s(n) = n + 1$, $\quad s : \mathbb{N} \to Y$.

$s(n) = n + 1$



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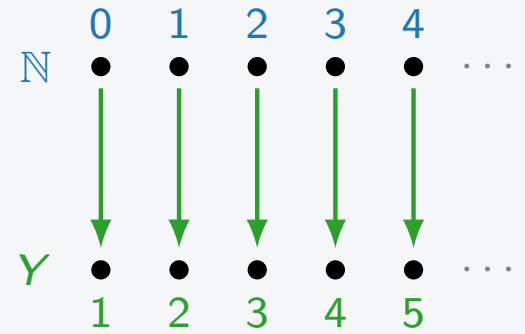
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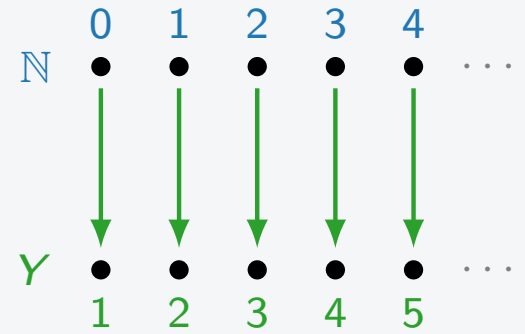
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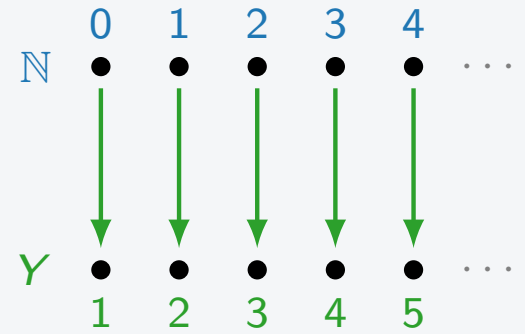
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- *injective*: different numbers stay different after adding 1
- *surjective*: every $m \in Y$ is reached, from $m - 1$

So s *is* a bijection from $\mathbb{N}$ to a proper subset: a counterexample.

Contradiction! ■

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Countably infinite sets: we can list them

Definition

A set X is **countably infinite** iff there is a **bijection** $f : \mathbb{N} \to X$.

$f(0), f(1), f(2), \ldots$ is a *list* of X: every element appears, and exactly once.

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n	0	1	2	3	$\cdots$
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$$f(n) = 2n + 1$$

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Countably infinite = listable: we can write the elements in an infinite list $x_0, x_1, x_2, \ldots$ that misses nothing and repeats nothing.

Also called *enumerable*. *Countable* means finite *or* countably infinite.

Strategy 3: proof by construction

> To show that a bijection **exists**, simply **build** one and check that it works.

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If A, B are countably infinite and disjoint, then $A \cup B$ is countably infinite.

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List A: $a_0, a_1, \ldots$ and B: $b_0, b_1, \ldots$

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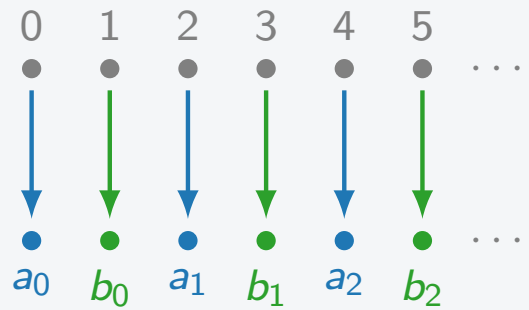
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List A: $a_0, a_1, \ldots$ and B: $b_0, b_1, \ldots$

Zip the two lists together:

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$h : \mathbb{N} \to A \cup B$



evens list A, odds list B

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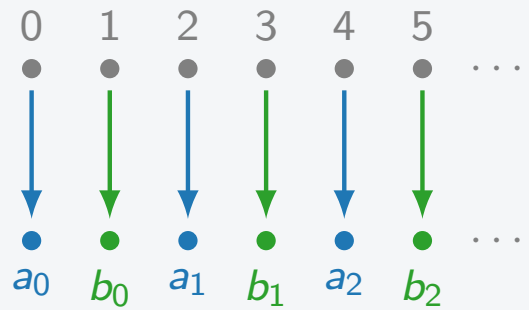
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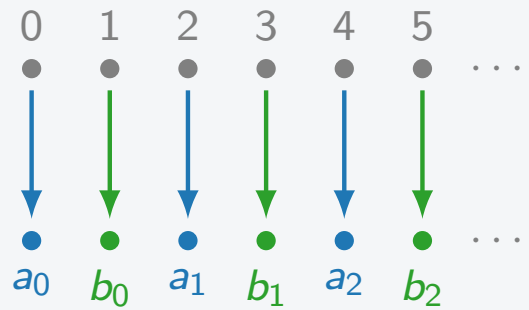
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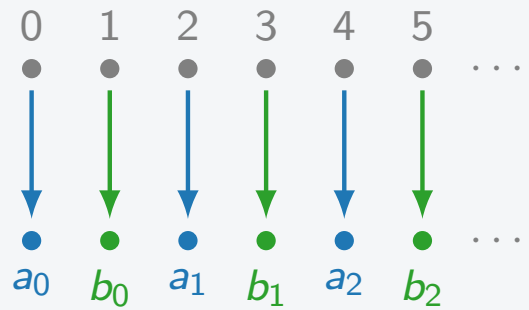
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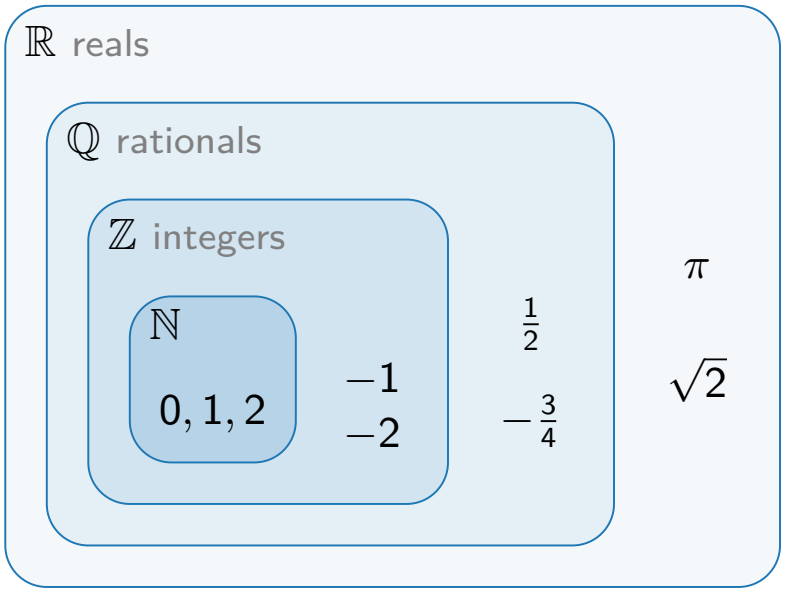
So this is a bijection $\mathbb{N} \to A \cup B$. ■

$h : \mathbb{N} \to A \cup B$



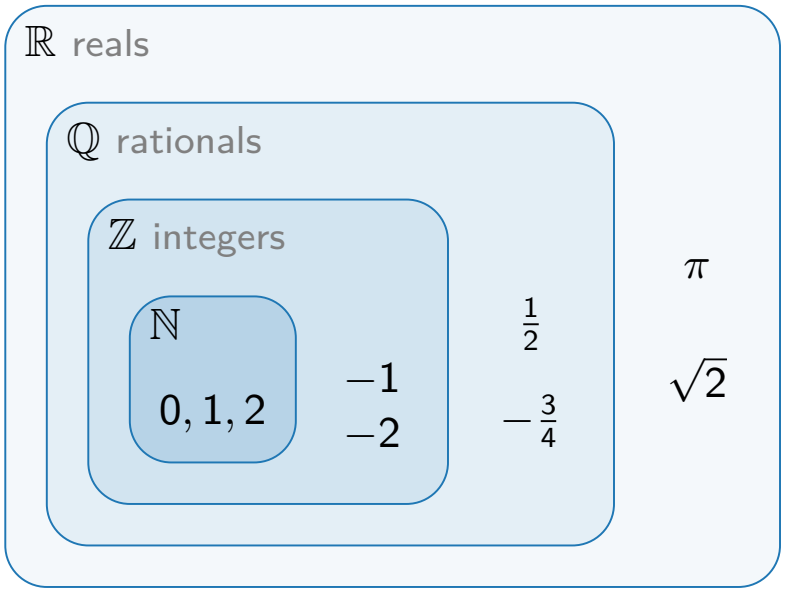
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Back to our question: is $\mathbb{R}$ larger than $\mathbb{N}$?



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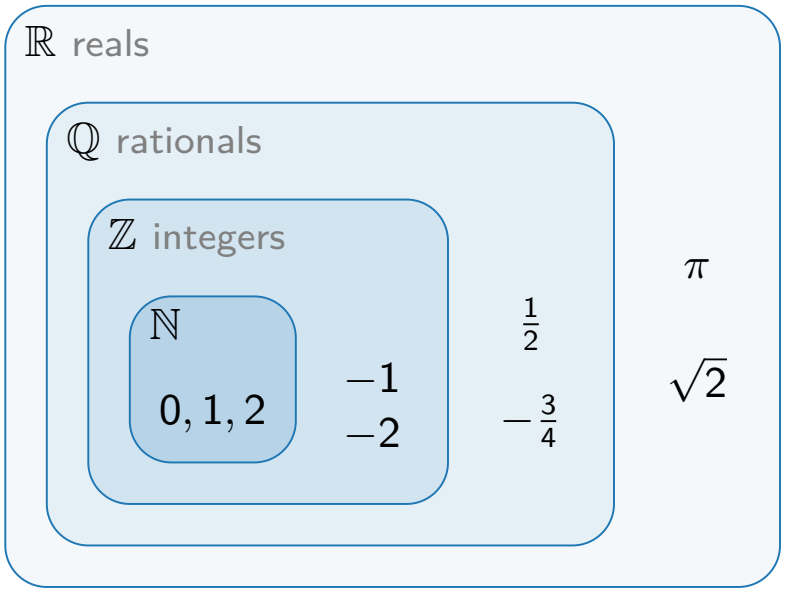
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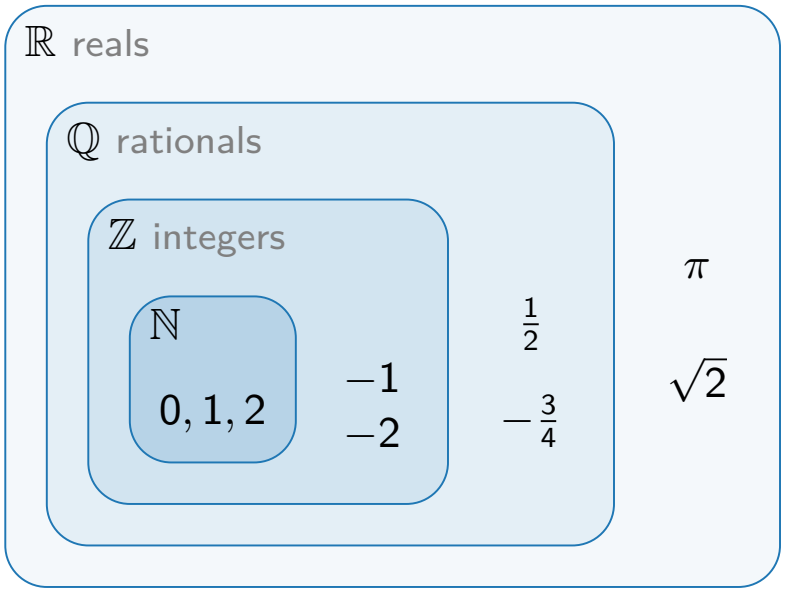


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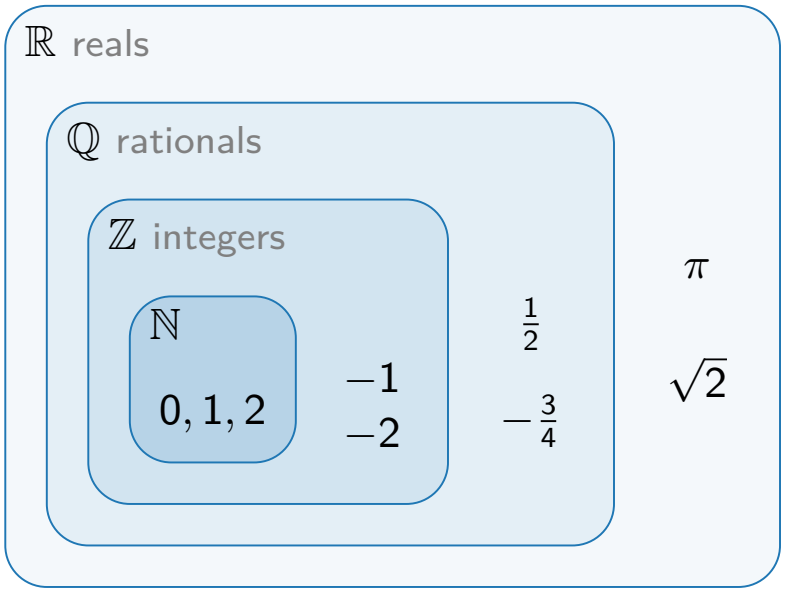
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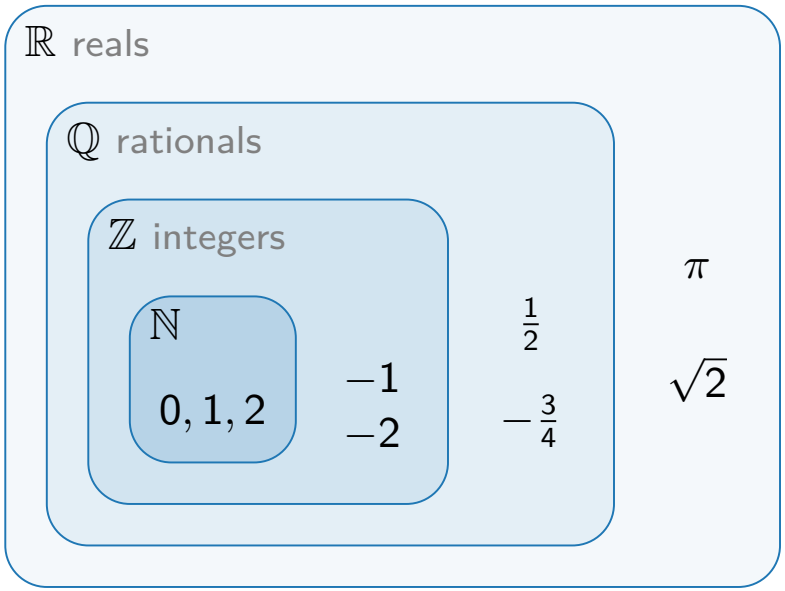
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uncountable: infinite, but not countably infinite. Cantor (1874)

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Cantor's theorem. For every set X: $|X| < |\mathcal{P}(X)|$.
So $|\mathbb{N}| < |\mathcal{P}(\mathbb{N})| < |\mathcal{P}(\mathcal{P}(\mathbb{N}))| < \cdots$ there are different infinities, there is no largest one

Interim summary: proofs and infinity

1. **Kinds of statements**

   accept definitions, assumptions

   prove theorems from them

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6. Sizes of infinity (advanced)

$\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ listable, $\mathbb{R}$ is not

Cantor: $|X| < |\mathcal{P}(X)|$

Quick check: proofs and infinity

True or false?

1. checking five examples proves a claim about all sets
2. a proof by contradiction starts by assuming the claim is false
3. $\mathbb{N} \setminus \{0\}$ is smaller than $\mathbb{N}$
4. $\emptyset$ is finite
5. the even numbers are countably infinite
6. all infinite sets have the same size

Quick check: proofs and infinity

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1. checking five examples proves a claim about all sets

   false
   examples can only refute

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   false
   $\mathbb{N}$ is countable, $\mathbb{R}$ is not

Propositional Logic

Languages have a syntax

A natural language: English

✓ *the cat sleeps*

✗ *cat the sleeps*

same words, but only one is a sentence

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what the well-formed expressions **mean**

Propositional logic (also *statement logic*) is a small formal language for *not*, *and*, *or*, *if*. Today: its syntax, then its semantics.

Building blocks: proposition letters and connectives

- Propositional logic models **propositions**: statements that are either **true** or **false**. (Not propositions: Is the earth round? Close the door!)

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Proposition letters (atoms)

$$\{P_a, P_b, P_1, P_2, \ldots\}$$

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Connectives

name	paraphrase	symbol
negation	"not"	$\neg$
conjunction	"and"	$\wedge$
disjunction	"or"	$\vee$
implication	"if . . . , then . . . "	$\rightarrow$
equivalence	"if and only if"	$\leftrightarrow$

Propositional formulas: a recursive definition

Definition: the formulas of propositional logic

(i) every proposition letter $P_a, P_b, \ldots$ is a formula

(ii) if φ is a formula, so is $\neg\varphi$

(iii) if φ and ψ are formulas, so are
$(\varphi \wedge \psi)$, $(\varphi \vee \psi)$, $(\varphi \rightarrow \psi)$, $(\varphi \leftrightarrow \psi)$

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Formula or not?

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To check a string, try to **build** it with rules (i)–(iv).
If there is no way to build it, it is not a formula.

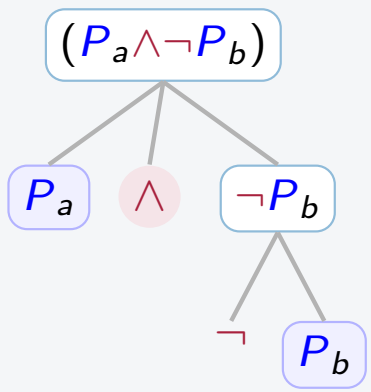
Syntactic trees: the structure of a formula

Each formula can be built in exactly **one** way. We draw that construction as a **tree**: each node is a subformula, the leaves are proposition letters.

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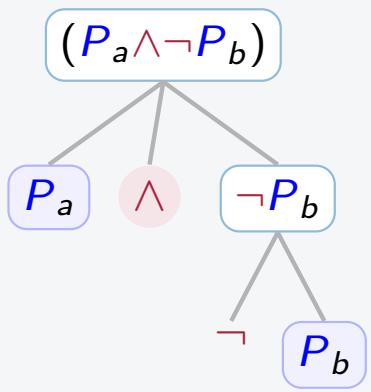
a conjunction



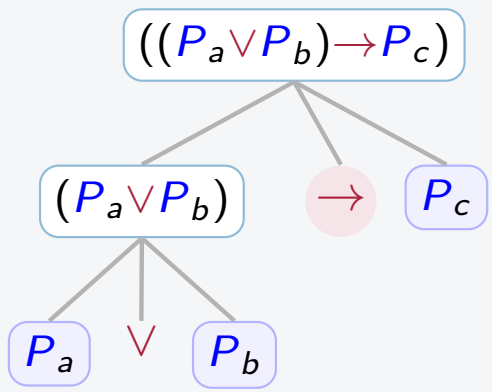
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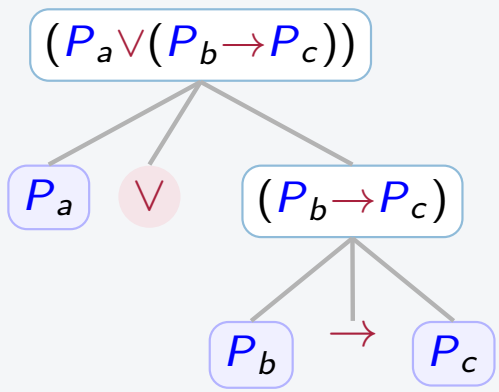
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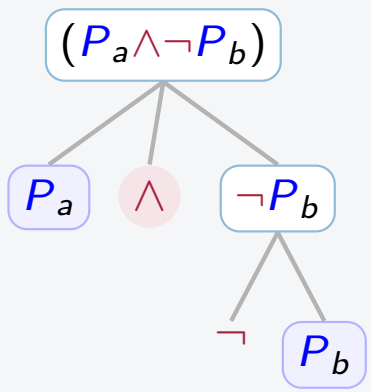
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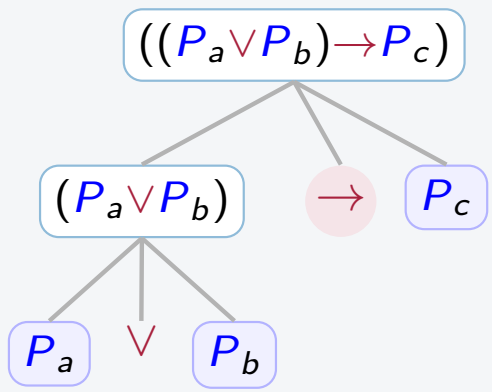
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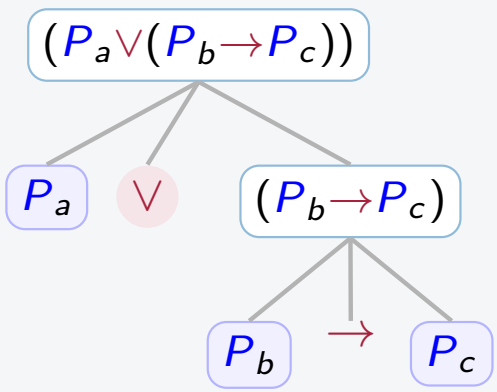
a conjunction



an implication



a disjunction



The last two use the same symbols in the same order, but have *different* trees: this is why brackets matter.

Some terminology

Kinds of formulas

- P_a **atomic**: a single letter
- $\ldots$ **complex**: anything else

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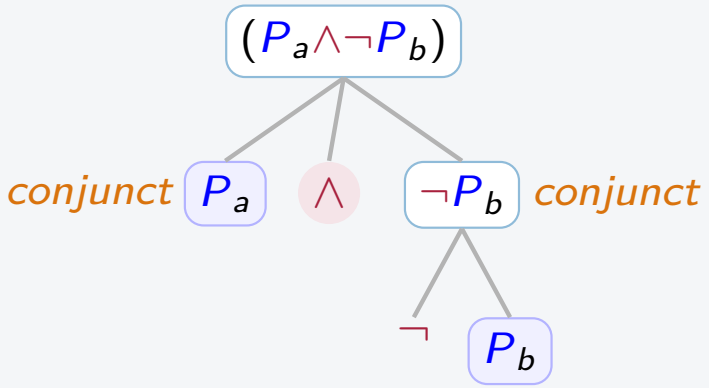
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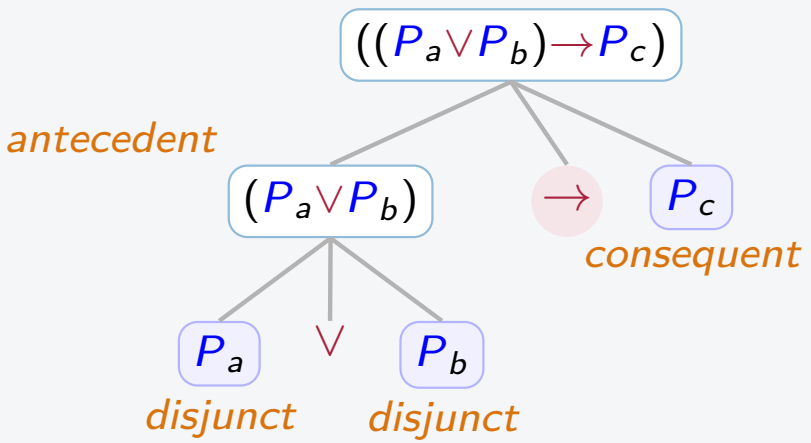
Names for the parts

$\varphi \wedge \psi$	*conjuncts*
$\varphi \vee \psi$	*disjuncts*
$\varphi \rightarrow \psi$	*antecedent* φ
	consequent ψ

a conjunction



an implication



Quick check: syntax

True or false?

1. $\neg\neg p$ is a formula
2. $p \wedge q \vee r$ is a formula
3. the main connective of $\neg(p \wedge q)$ is $\wedge$
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5. in $p \rightarrow q$, p is the antecedent
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   rule (ii), twice

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Go through these carefully!

Quick check: translation

Our dictionary

P_r: *it rains* P_w: *the street is wet* P_c: *class is cancelled*

Translate!

1. $\neg P_r$
2. $(P_r \wedge \neg P_w)$
3. $(P_r \rightarrow P_w)$
4. $\neg(P_r \wedge P_w)$

5. *if it rains, class is not cancelled*
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5. *if it rains, class is not cancelled* — $(P_r \rightarrow \neg P_c)$
6. *it does not rain, but the street is wet* — $(\neg P_r \wedge P_w)$ *but* is $\wedge$ for logic

Compare 4 and 6: brackets matter!

Semantics: what do formulas mean?

Semantics: truth values

- Every formula is built from **atomic** propositions $P_a, P_b, \ldots$

Semantics: truth values

- Every formula is built from **atomic** propositions $P_a, P_b, \ldots$
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P_e *the earth is round* **1**

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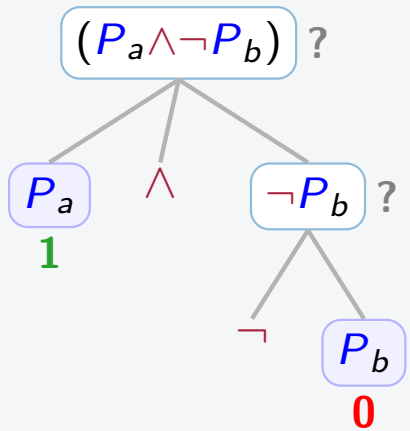
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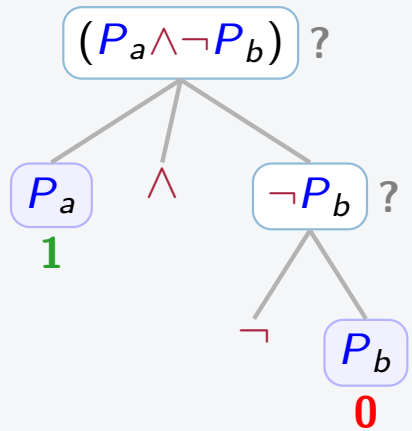
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Each **connective** is a **function** on truth values (lecture 2)

$$\neg : \{0,1\} \to \{0,1\} \qquad \wedge : \{0,1\} \times \{0,1\} \to \{0,1\}, \quad \ldots$$

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A new tool: truth tables

φ	ψ	$\varphi \wedge \psi$
1	1	**1**
1	0	**0**
0	1	**0**
0	0	**0**

left columns	the values of the *parts*
right column	the value of the *whole*
one row	one possible situation
all rows	*every* combination

A truth table lists a connective's function, one row per input. With 2 parts there are $2 \times 2 = 4$ rows; with n parts, 2^n (just like $|\mathcal{P}(X)| = 2^{|X|}$).

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The meaning of the connectives

One truth table for all five connectives

φ	ψ	
1	1	
1	0	
0	1	
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Sound familiar? The same words defined the set operations

$X \cap Y = \{z \mid z \in X \textbf{ and } z \in Y\}$ $\quad \overline{X} = \{z \in U \mid \textbf{not } z \in X\}$

$X \cup Y = \{z \mid z \in X \textbf{ or } z \in Y\}$

Conjunction: *and*

- Example: $(P_r \wedge P_w)$ *it rains and the street is wet*

One row per situation

	P_r	P_w	$P_r \wedge P_w$
s_1	1	1	
s_2	1	0	
s_3	0	1	
s_4	0	0	

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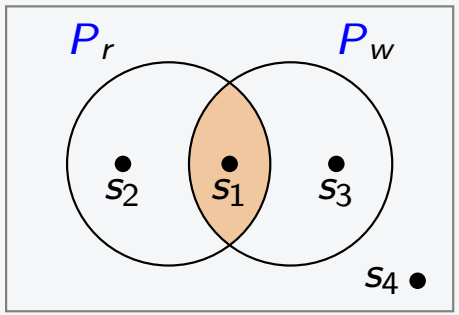
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In a picture



each dot is a situation
shaded (formula is 1):
$\{s_1\}$ = the *intersection*
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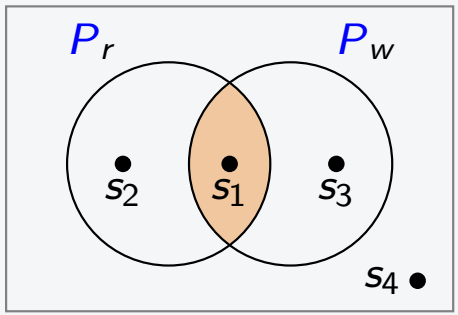
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$\wedge(x, y) = 1$ exactly when *both* $x = 1$ and $y = 1$
and, *but*, *although*: all become $\wedge$ in logic

Disjunction: *or*

- Example: $(P_r \vee P_w)$ *it rains or the street is wet*

One row per situation

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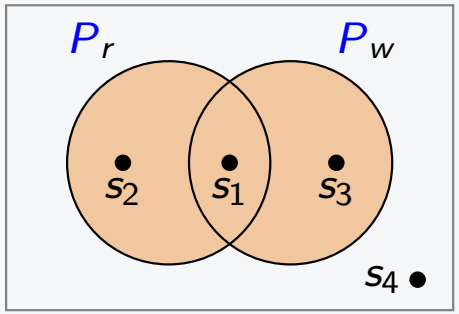
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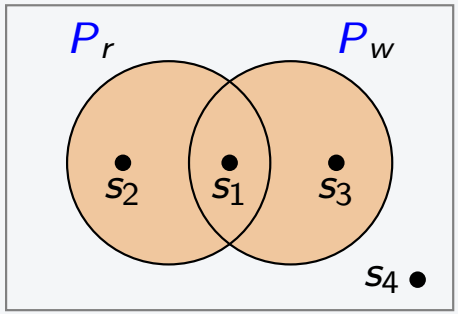
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$\vee(x, y) = 1$ exactly when *at least one* of x, y is 1
inclusive: both may be true (unlike "soup or salad?" on a menu)

Implication: *if . . . , then . . .*

- Example: $(P_r \rightarrow P_w)$ *if it rains, then the street is wet*

One row per situation

	P_r	P_w	$P_r \rightarrow P_w$
s_1	1	1	
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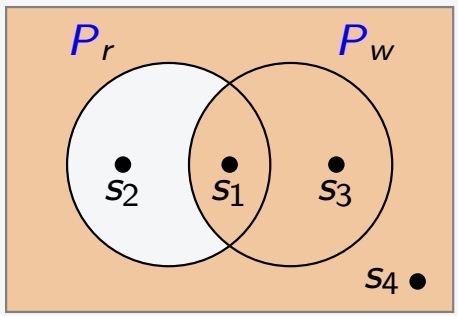
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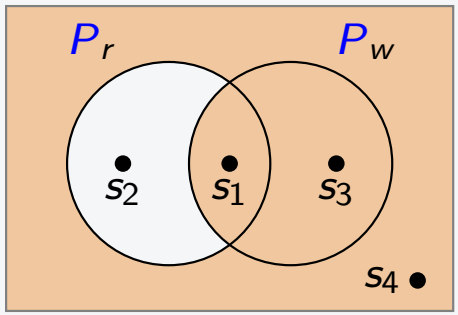
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- Example: $(P_r \to P_w)$ *if it rains, then the street is wet*

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	P_r	P_w	$P_r \to P_w$	
s_1	1	1	**1**	rain, wet street
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$\to(x, y) = 0$ exactly when $x = 1$ and $y = 0$
if the antecedent is 0, the implication is 1 (recall the promise!)

Equivalence: *if and only if*

- Example: $(P_r \leftrightarrow P_w)$ *it rains if and only if the street is wet*

One row per situation

	P_r	P_w	$P_r \leftrightarrow P_w$
s_1	1	1	
s_2	1	0	
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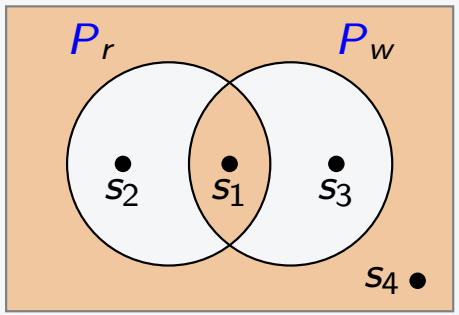
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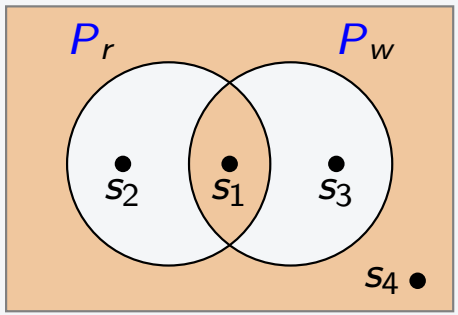
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$\leftrightarrow(x, y) = 1$ exactly when $x = y$
an implication in *both* directions: $P_r \rightarrow P_w$ and $P_w \rightarrow P_r$

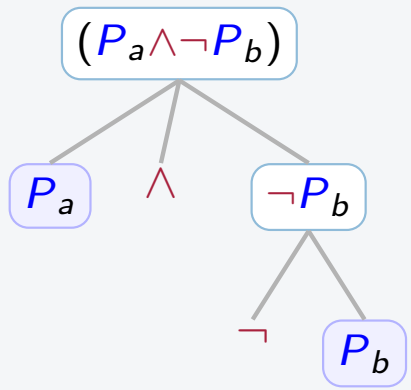
Looking ahead: truth tables for complex formulas

- Build the table **bottom-up**, one column per subformula:
  P_a, P_b $\rightsquigarrow$ $\neg P_b$ $\rightsquigarrow$ $(P_a \wedge \neg P_b)$

The truth table

	P_a	P_b		
s_1	1	1		
s_2	1	0		
s_3	0	1		
s_4	0	0		

The tree



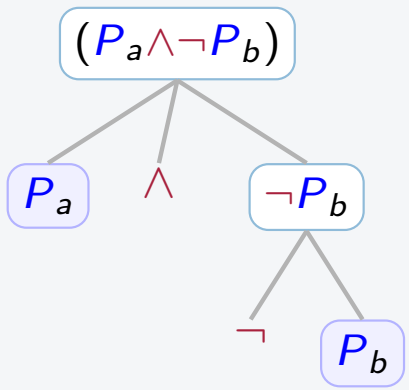
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The truth table

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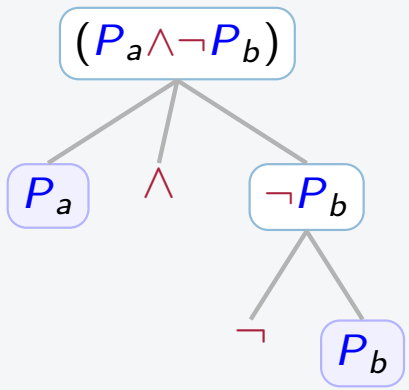
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s_4	0	0	**1**	**0**

The tree



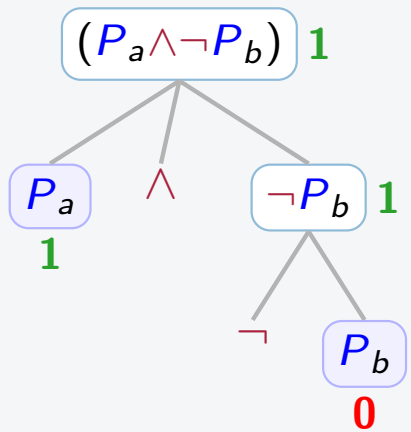
Looking ahead: truth tables for complex formulas

- Build the table **bottom-up**, one column per subformula:
  P_a, P_b $\rightsquigarrow$ $\neg P_b$ $\rightsquigarrow$ $(P_a \wedge \neg P_b)$

The truth table

	P_a	P_b	$\neg P_b$	$(P_a \wedge \neg P_b)$	
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s_2	1	0	**1**	**1**	$\leftarrow$
s_3	0	1	**0**	**0**	
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Row s_2, on the tree



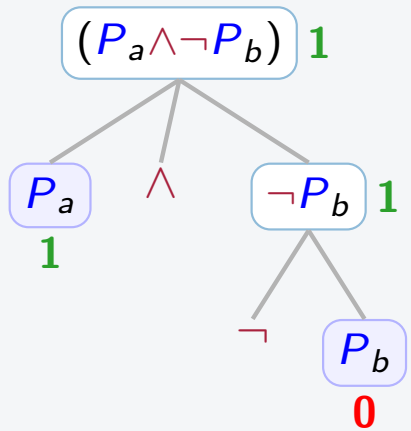
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Next week: this method works for *any* formula. We will use it to find formulas that are *always* true, *never* true, or that *mean the same*.

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Next time: truth tables for complex formulas.